

Vennela Neelap

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EXPERIENCE

Associate Software Engineer

September 2025 – Present

Dr. Paul Fisher & Associates

- Developed a production C#/.NET Worker Service integrating DaySmart and RevolutionEHR REST APIs through OAuth 2.0; used Swagger/OpenAPI to validate endpoints, payload schemas, authentication flows, and responses.
- Implemented configurable polling, incremental synchronization, historical backfills to 2020, overlap windows, pagination, duplicate prevention, retry handling, and structured logging for reliable patient and appointment data exchange.
- Deployed the integration as a Windows Service and supported firewall, network, and remote-access configuration; developed Tableau views of patient flow, appointment trends, and accounting data while maintaining HIPAA-conscious data handling.

Software Engineering R&D Intern

May 2025 – August 2025

Honeywell

Dallas, TX

- Configuration Management and Validation Platform: Engineered and deployed an end-to-end configuration-management application using C#, .NET, Blazor, ASP.NET Core, and SQL Server to upload and validate JSON, XML, and TXT manufacturing files, supporting an estimated 100+ configurations per month.
- Designed secure REST APIs using ASP.NET Core, Swagger/OpenAPI, RBAC, and SQL Server; implemented schema validation, structured error handling, and audit-ready access controls for manufacturing configuration data.
- Built automated validation and verification tests with NUnit and TUnit Selenium, performing root-cause analysis on API and database-integration defects and reducing validation time by an estimated 60%.
- Equipment and Pressure-Plate Accountability System: Developed an accountability system using C#, .NET, SQL Server, and LabVIEW to track an estimated 50+ weekly equipment and pressure-plate checks across MEMS and sensor-fabrication operations.
- Contributed to hardware/software system design with LabVIEW, .NET, and SQL Server, connecting application workflows with fabrication equipment requirements and supporting end-to-end operational automation.
- Created test-automation and verification workflows using LabVIEW and .NET; investigated failed checks through root-cause analysis and reduced manual accountability gaps by an estimated 40%.
- Material Cost and Pricing Automation: Automated the material-pricing workflow end to end using C#, .NET 6, ODBC, SQL Server, EPPlus, and CsvHelper, processing an estimated 1,000+ material records per run from Excel/CSV inputs.
- Engineered material-matching, input-validation, and update logic with C#, ODBC, and SQL Server; added exception handling and transaction-aware processing to protect pricing accuracy during bulk updates.
- Implemented automated verification and structured JSON logging with System.Text.Json, reducing an estimated four-hour manual process to under 15 minutes while providing an audit trail for failed records and root-cause analysis.
- Delivered all three projects through a Jira- and Git-based Agile workflow within a broader .NET, AWS, and J2EE engineering environment, communicating progress, technical risks, requirements, and next steps across engineering and manufacturing teams.
- Regularly presented project milestones, system designs, application demonstrations, validation results, and ROI to a director and cross-functional stakeholders; created technical slide decks that translated complex software and hardware workflows into actionable business updates.
- Led recurring stakeholder reviews, incorporated feedback into project priorities, and communicated engineering decisions to both technical and nontechnical audiences; received a Honeywell Bravo Award for technical contributions and cross-functional collaboration.

PROJECTS

SkillSharp AI

October 2024 – Present

Full-Stack Generative AI Application

- Designed and deployed a generative-AI interview platform using Gemini LLM and prompt engineering to generate contextual technical questions and structured feedback for 30+ users.
- Engineered a scalable full-stack architecture with Next.js, React, TypeScript, PostgreSQL, and Drizzle ORM, integrating RESTful services and Swagger/OpenAPI contracts.
- Implemented Clerk authentication, session management, protected routes, and role-aware data access; iteratively evaluated prompts to improve relevance, output consistency, and hallucination resistance.

Collaborative Document Editor

April 2025 – June 2025

Real-Time Backend Systems

- Architected a real-time collaborative editing service with NestJS, MongoDB, and WebSockets, using operational transformation to reconcile simultaneous document changes.
- Supported 25+ concurrent users through event-driven bidirectional communication, persistent document state, and conflict-aware synchronization.

LibraTrack

January 2025 – February 2025

Full-Stack Library Management Platform

- Developed and deployed a full-stack library-management platform using React, Spring Boot, PostgreSQL, and AWS S3, hosting the frontend on Vercel and backend on Render.
- Secured RESTful APIs with JWT authentication, HTTP-only cookies, RBAC, and protected routes; integrated persistent relational data and S3 object-storage workflows.

InvestIQ

January 2025 – February 2025

Venture and Investment Analytics

- Engineered a full-stack venture and investment analytics platform with FastAPI and React, aggregating multiple external data APIs into a unified decision-support interface.
- Reduced API response latency by approximately 40% by optimizing backend request orchestration, response processing, and frontend/backend data flows.

BankApp Platform

August 2026 – Present

C# Console and REST API (In Progress)

- Designed an object-oriented C#/.NET banking domain with abstract accounts, interfaces, inheritance, encapsulation, checking/savings rules, and validated deposit, withdrawal, and transfer workflows.
- Extended the console application into an ASP.NET Core REST API, implementing customer and transaction endpoints, request models, dependency injection, and Swagger/OpenAPI documentation.
- Added a MongoDB repository layer for persistent customer/account data and used Git feature branches to separate console, no-database API, and MongoDB API development.

CineLog Open-Source Contribution

March 2026 – July 2026

Python / Flask Testing

- Refactored a Flask/SQLAlchemy watchlist service to use UUID film identifiers, modern database retrieval, duplicate-entry prevention, and explicit FilmNotFoundError handling.
- Authored Pytest fixtures and regression tests for invalid-film scenarios; resolved six code-review comments, rebase conflicts, and history updates with all five tests passing.

PathReview Open-Source Contribution

AI Engineering / Testing

- Diagnosed an unhandled missing-document edge case in an AI-assisted review service and implemented descriptive failure handling for profiles without ingested content.
- Added two regression tests aligned with existing FastAPI, SQLAlchemy, and Pytest patterns; validated the change against a 377-test passing suite.
- Completed the open-source delivery lifecycle through issue reproduction, technical documentation, Git branching, rebasing, conflict resolution, code review, and pull-request submission.

BookSmart

November 2024 – December 2024

Full-Stack Book Management System

- Engineered a layered full-stack application with ASP.NET Core, React, Entity Framework Core, and SQL Server, separating presentation, API, business-logic, and persistence concerns.
- Designed Swagger-documented CRUD REST APIs, relational entity models, server-side validation, and responsive client workflows for reliable database-backed record management.

ECG-FM Parameter-Efficient Fine-Tuning

Healthcare AI / Deep Learning

- Parameter-efficiently fine-tuned an ECG foundation model on 65,424 signals using per-sample normalization, stratified train/validation/test splits, and a reproducible 15-epoch training pipeline.
- Trained only 2.60% of model parameters while achieving 0.9397 validation AUC and 0.9308 test AUC, demonstrating strong generalization with reduced compute requirements.
- Improved validation F1 from 0.4659 to 0.7551 and accuracy from 0.5670 to 0.7589; tracked precision, recall, loss, and AUC for model evaluation.

Text-Conditioned Adv-Diffusion

Generative AI / Computer Vision

- Reproduced and extended an adversarial latent-diffusion pipeline with Stable Diffusion v1.5, DDIM sampling, and face-parsing masks to generate localized, text-conditioned facial perturbations.
- Integrated LLaVA-1.5 7B semantic descriptions with CLIP alignment, ArcFace identity, and LPIPS perceptual objectives, balancing multimodal alignment, visual fidelity, and identity robustness.
- Optimized GPU memory through model placement and CPU offloading on a 12 GB RTX system; increased FFHQ text-image similarity from approximately 0.18-0.24 to 0.36-0.52 after attack.

Credit Card Fraud Detection

August 2024 – Present

Machine Learning / Financial Analytics

- Built a reproducible fraud-analytics pipeline for more than 1 million financial transactions, performing data cleaning, exploratory data analysis, feature inspection, and visualization with Pandas, NumPy, and Matplotlib.

- Trained and evaluated a logistic-regression classifier for an imbalanced dataset, emphasizing precision, recall, F1, and confusion-matrix analysis over misleading accuracy alone.

CycleGAN Image-to-Image Translation

Generative AI / Computer Vision

- Implemented an unpaired image-to-image translation pipeline in PyTorch using dual generators, discriminators, adversarial loss, and cycle-consistency objectives.
- Built repeatable training and inference workflows to evaluate translation quality, domain consistency, convergence behavior, and common GAN failure modes.

Decoder-Only Language Model

Natural Language Processing

- Built a decoder-only Transformer language model in PyTorch, implementing tokenization, embeddings, positional information, causal self-attention, and autoregressive decoding.
- Developed training and text-generation pipelines with next-token prediction, loss monitoring, and qualitative output evaluation across iterative model experiments.

SmartCandlestick

C++ Market-Data Application

- Developed a C++ market-data processing application using object-oriented design, reusable candlestick models, STL data structures, and structured file ingestion.
- Aggregated price observations into open-high-low-close candlesticks for console-based trend inspection, strengthening algorithmic processing and financial-data modeling.

InCollege Platform

August 2024 – 2025

COBOL Networking and Job Portal

- Collaborated in Agile sprints to build a modular COBOL professional-networking platform supporting authentication, profiles, connections, messaging, job postings, and applications.
- Implemented input validation, record search, sequential .dat file persistence, and structured PIC/OCCURS data models; containerized the development environment with Docker.

Portfolio Website

Frontend Development

- Designed and deployed a responsive personal portfolio using semantic HTML, CSS, and JavaScript to showcase 12+ software, AI, and academic projects.
- Implemented accessible navigation, responsive layouts, and interactive project content, reaching more than 500 unique visitors.

SKILLS

Languages: Python, C#, Java, C++, JavaScript, TypeScript, .NET
Frameworks: ASP.NET Core, Blazor, .NET Worker Services, React, Next.js, Node.js, NestJS, FastAPI, Spring Boot, Flask
Cloud & DevOps: AWS, Docker, Kubernetes, GitHub Actions, GitLab CI/CD, Jenkins, Terraform, Linux
Databases: PostgreSQL, SQL Server, MySQL, MongoDB, Entity Framework Core, Drizzle ORM, SQLAlchemy
AI/ML & Analytics: PyTorch, Hugging Face, Diffusers, Gemini AI, Pandas, NumPy, Matplotlib, Tableau

EDUCATION

University of South Florida	Tampa, FL
<i>Bachelor of Science in Computer Science.</i>	<i>May 2026</i>